

Secure Exit



SIMATS Online Programming Lab

JAVA PROGRAMMING



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QUESTIONS

Q1

Banking Transaction System –
Exception Handling

10 marks

Q2

Hospital Patient Registration –
User-Defined Exception

10 marks

Q3

E-Commerce Order Processing
– Built-in Exceptions

10 marks

Q4

Hospital Emergency System –
Thread Priority

10 marks

Q5

Online Food Delivery – Multiple
Threads and Execution Order

10 marks

Q3 E-Commerce Order Processing – Built-in Exceptions

10 marks

Develop a Java program for an online shopping application. Store product names and prices in arrays and accept a product index and quantity from the customer.

The program must:

- Calculate total cost as quantity × price.
- Handle invalid product indices.
- Handle invalid numeric input.
- Reject zero or negative quantities.
- Use appropriate built-in exception classes.
- Continue execution after handling an invalid transaction.

Sample Test Cases:

Input / Scenario Expected Result Purpose

1, 2 Valid product; total cost calculated Normal case

5, 2 `ArrayIndexOutOfBoundsException` handled Boundary / exception case

2, -1 Invalid quantity Invalid input

abc, 2 `InputMismatchException` handled Invalid input

Your Code

Input

5/5 answered

```
1 class Main
2 {
3     public static void main (String[] args)
4     {
5         int index = 1, qty=2;
6         double [] price={100,200,300};
7         try
8         {
9             if(qty<=0)
10            {
11                throw new IllegalArgumentException(
12            }
13            System.out.println("Total : "+price
14        }
15        catch(ArrayIndexOutOfBoundsException e)
16        {
17            System.out.println("Invalid product ind
18        }
19        catch(IllegalArgumentException e)
20        {
21            System.out.println(e.getMessage());
22        }
23    }
24 }
```

stdin input

Output

Total : 400.0

✓ Submitted